

CSC111 Lab
Object Oriented Programming – Lab III
---- Lab 9 ----

Create a project **Lab09**

Q1) Write a Java class **Car** that represent a car in a car dealership. The car has the following attributes:

- **model**: String.
- **year**: int. The year the car was made.
- **mileage**: int representing how many kilometers the car has drove.
- **soom**: double. Best offer for the car so far.
- **hadd**: double. Asking sale price. Must be greater than soom.

The class has the following methods:

- **Car()**
- **Car(String model, int year, int mileage, double soom, double hadd)**
- **setters and getters for model, year, mileage, soom and hadd.**
- **similar(Car c):boolean** Compares two objects, the calling object and the argument c and returns true if they are similar, false otherwise. Two cars are similar if they have same model, year and within 10k of eachother mileage. For example, 2014 ford with mileage 120K is similar to a 2014 ford with mileage 125K.
- **difference():double.** the difference between soom and hadd. Must be positive.
- **printCar(): void**

Car
- model: String - year: int - mileage: int - soom: double - hadd: double
+ Car () + Car(String model, int year, int mileage, double soom, double hadd) + setModel(String model):void + getModel():String + setYear(Int year):void + getYear() :Int + setMileage(int mileage):void + getMileage() :int + setSoom(double soom):void + getSoom() :double + setHadd(double had):void + getHadd():double + similar(Car c): boolean + difference():double + printCar(): void

Write a class **testCar** that will do the following:

1. Declare objects **car1** and **car2** of class Car.
2. Read car attributes from the user using the both constructors.
3. Print both cars information.
4. Check if both cars are similar.
5. Ask the user repeatedly to enter soom for car1 until **soom** is within 2% of **hadd**.
6. Print both cars information.
7. Feel free to try creating additional objects, read their data and manipulate them.
8. exit.

Sample Run

Please enter car1 model, year, mileage, hadd and soom

Ford 2018 200000 100000 80000

Please enter car2 model, year, mileage, hadd and soom

Toyota 2019 200000 300000 250000

This car is a Ford and was made in 2018.

It has 200000 KM and soom is 80000.0 Hadd is 100000.0

This car is a Toyota and was made in 2019.

It has 200000 KM and soom is 250000.0 Hadd is 300000.0

Cars are not similar

Car1 difference between soom and hadd must be less than 2000.0

The difference is = 20000.0 SR

Please enter another soom> 85000

The difference is = 15000.0 SR

Please enter another soom> 98000

The difference is = 2000.0 SR

Please enter another soom> 98001

The difference is = 1999.0 SR which is less than 2000.0